Generative AI and Science: a short primer (to read before the transcript)

Authored by Carsten Bergenholtz and ChatGPT-5.2 Pro (13.02.26). I provided the structure, all topics and points, as well as the overall learning goal of the text. ChatGPT-5.2 Pro wrote up the text, based on this input. The writing process involved about 30 prompts, to modify flaws and misguided drafts proposed by ChatGPT-5.2 Pro.

Introduction

This note is a short introduction to what generative AI (GenAI) is, what kind of knowledge it can produce, and which ethical issues it raises. GenAI usually refers to text-based chatbots built on large language models (LLMs), but note that products such as ChatGPT or Google Gemini also can produce images or even videos. The transcript you will read (subsequently) assumes some of these ideas. The goal is not technical mastery. The goal is to help you think clearly about what GenAI tools can help you do, when they are likely to mislead, and what that implies for doing good science.

1) What is GenAI?

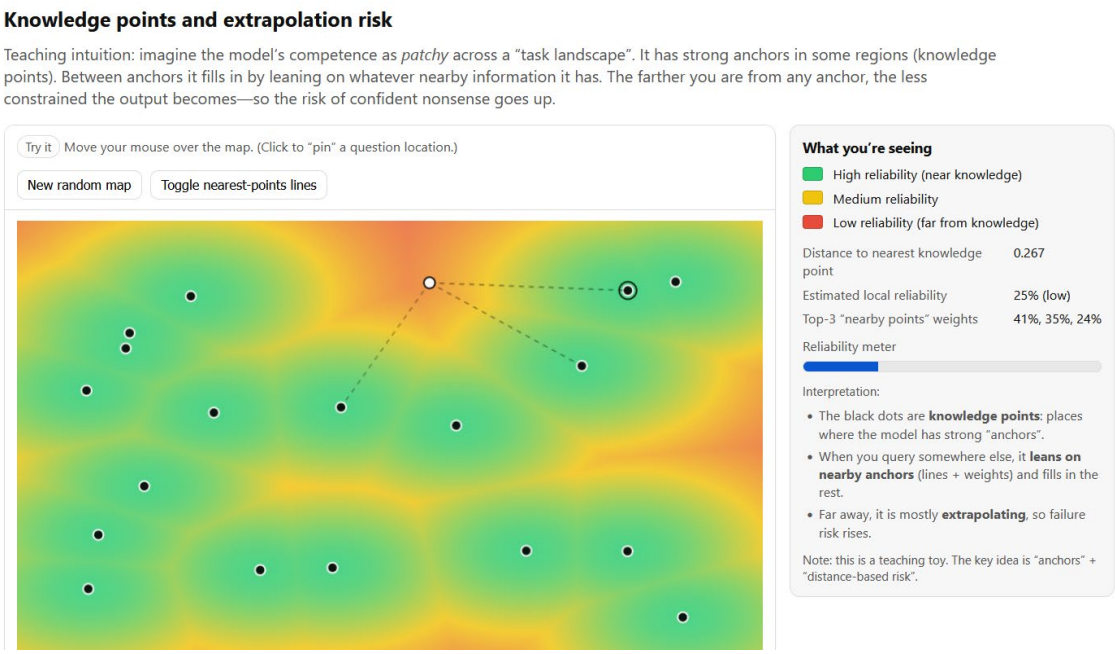
A useful first intuition is that LLM-based GenAI mainly learns statistical patterns in language: which words and ideas tend to appear together, and in what order. It does not start with a built-in model of the social world (firms, incentives, motives) or the physical world (chemistry, biology). It starts with patterns in data.

To illustrate: if you ask a GenAI chatbot to "pretend to be a doctor" and then request it tells you its favorite song, it is likely to answer Bon Jovi's "Bad Medicine." Not because doctors have that particular preference, but because the words "doctor" and "medicine" are frequently linked in the data it has seen. That is the core point: GenAI is designed to associate words and producing plausible continuations.

How does a model learn these associations? In pretraining, it is trained on massive datasets (books, articles, websites, code, etc.) and repeatedly asked to predict the next word/token in a text. With enough data and compute, this becomes much more than standard autocomplete: the model learns to produce long, coherent text and to imitate many styles of writing. But its competence is still shaped by what it has been trained on. This does not mean that it can only repeat what it has already seen – GenAI can generate output (texts, images, videos) that the world has never seen before. Yet, GenAI performs the best within familiar territory—problems and text that resemble what it has seen a lot of during training. François Chollet describes this as "local generalization": strong performance on 'known unknowns' (variations of familiar patterns), and weaker performance when tasks are genuinely out-of-distribution or require domain insight that the model has not effectively learned (Chollet 2019, see also https://x.com/fchollet/status/1847790550067663102). Hence, if you 'push' GenAI to combine two closely related topics or areas it will perform better than if you expect it to go into completely unknown territory. Put differently, if you trained a LLM on all scientific documents before a genuinely novel scientific discovery, they (so far, February

26) have been unable to make such genuine leaps. The major field-changing discoveries that have been attributed to LLMs, have generally involved humans framing the topic, questions and workflow, as well as verifying the output.

The following visualization aims to make the point. The closer to the green areas (near lots of information processed) the model is fairly reliable. The more we move into territory where the model has processed no or little information (orange/red), the responses become much more unreliable.



https://carstenbergenholtz.github.io/Phil-of-Science-tools/GenAI_knowledge_landscape_analogy.html

Also keep in mind that GenAIs are probabilistic, which means that whatever result you get in one instance, might not be the same you get immediately afterwards, with the exact same prompt.

Nevertheless: Modern chatbots feel much more capable than "next-token prediction" suggests because they are not just pretrained models that guess a next token (≈ word).

- RLHF (Reinforcement Learning from Human Feedback): human raters compare and rank alternative answers provided by chatbots. These rankings train a 'reward model' of what humans prefer, and the system is optimized to score higher on that reward. Hence, GenAI learns which answers humans tend to prefer.
- Reasoning/problem-solving training: in some domains (especially math, science problems, and coding), systems can also be trained with an automated system that offers feedback on whether the output is right or wrong – e.g. when building code, that can be immediately verified.
- Tool use: in many setups the model can browse, retrieve documents, run code, or use other tools. Tool use can improve reliability because the model can check facts, test

code, and update answers based on retrieved sources. But it also creates new failure modes (e.g., picking the wrong source, misunderstanding a source, or summarizing it incorrectly).

GenAI systems are showing rapid improvements on a range of benchmarks, have anecdotally been shown to play a vital role in knowledge production (e.g. in science), and recent AI systems have shown competition-level performance in areas like mathematics and programming: there are 'Olympiad-style' competitions where these systems have been able to match the performance of some of the best human participants (Luong & Lockhart 2025).

It does seem like GenAI performance is best in certain domains; where tasks have clear success criteria and strong feedback signals, so systems can be trained and evaluated in a more disciplined way. I expand on this in the following.

2) Why GenAI tends to excel on well-defined tasks (and why many research tasks are "messy")

Well-defined tasks have a clear target and a clear scoring rule—so both the system (during development) and the user (during use) can quickly tell what "good" looks like. In math, an answer is correct or incorrect. In programming, code either runs and passes tests or it doesn't. By trying out different answers, and getting feedback on correct answers, the GenAI models can learn from these feedback loops, which improves their capabilities.

Many research tasks, by contrast, are "messy": the goal is partly undefined, the context matters, and there often isn't a single right answer or an immediate way to test correctness (see e.g. Garicano 2026; https://www.siliconcontinent.com/p/a-new-years-letter-to-a-young-person). This means it becomes much more difficult to prepare the GenAI models for these kinds of less well-defined and more unfamiliar problems.

Tool use can sometimes reduce messiness by turning parts of the task into checkable sub-tasks (e.g., browse to verify a factual claim, retrieve sources, run code to test a result). But the hard part often remains: deciding what to trust, what to doubt, and what "good science" requires in that specific context.

These insights also relate to what status the knowledge and explanations that GenAI models produce arguably have, as outlined in the following section.

3) Epistemology: causality vs. understanding (Verstehen)

In this course we distinguish between two broad ideals of explanation (see Bergenholtz 2026f). One ideal emphasizes causality: identifying what causes what, why it happens, and what would change if we intervened (counterfactual thinking). Another ideal emphasizes understanding (Verstehen): making sense of human meanings, intentions, and context, and producing interpretations that are credible and insightful. GenAI can generate text that looks like both kinds of explanation. But remember the 'Bad Medicine' intuition from earlier: it is often producing plausible language based on learned patterns. So a good-sounding explanation is not automatically a good causal account. Furthermore,

if one asks the GenAI model why it provided the given explanation, its reasoning does not necessarily reflect the actual underlying computational processes.

For causal research, GenAI can be genuinely useful in several stages: mapping an existing literature quickly, generating candidate hypotheses, suggesting alternative causal stories to consider, drafting analysis plans, and writing code for data cleaning and statistical analysis. Used well, it can speed up the 'implementation' part of research.

For Verstehen-oriented research, GenAI can help with practical tasks (drafting interview guides, proposing tentative codes, summarizing text, comparing themes across documents). But Verstehen (in Weber's sense) is not just "producing an interpretation"; it is understanding social action from the actor's point of view—what people mean, intend, and experience in a specific context. A GenAI system cannot literally take that perspective. It has no intentions, no stakes, and no lived/embodied experience of the setting; it only has learned patterns from text. That means it can imitate an interpretive voice without actually grasping what is meaningful to participants in that situation. As a result, AI-generated interpretations can easily become generic, culturally stereotyped, or overly confident and "too tidy." In Verstehen work, GenAI output is best treated as prompts or candidate readings to test against the material, not as an interpretation to accept. Human researchers still need to do the interpretive work: engage with context, be reflexive about assumptions, consider alternative interpretations, and justify claims by showing how they follow from concrete data (quotes, episodes, field notes) rather than from plausible-sounding narrative.

A final point (often undervalued): expert users get more value and take fewer risks. Researchers like Andy Hall and Joshua Gans (who feature in the subsequent transcript) are domain experts and experienced GenAI users. They know what a good answer should look like, can detect subtle mistakes, and can set constraints (e.g., what counts as evidence, which checks to run, when to distrust an output). Non-experts are much more likely to accept fluent output as truth—especially on messy tasks.

4) Ethics: Integrity, Transparency and Copyright

Plagiarism and Authorship

A key requirement (at Aarhus University) is the use of a generative AI declaration. While it may be challenging to exactly specify if knowledge or text originated in the researcher's own reasoning or through / from GenAI assistance, it is nevertheless important an important step toward honesty and transparency to distinguish between such sources. Also keep in mind that even state-of-the-art GenAI models (still) hallucinate – in particular in continuous iterations where the chatbot is to build upon prior dialogue (Fan et al. 2026). Hence, references, quotes and details should always be verified..

Cultural Bias in Ethical Judgments

Whatever is in the training data—and the choices made during model development—can show up in GenAI outputs. If the training data contains systematic patterns or stereotypes (e.g., associating certain jobs with particular genders or ethnicities), the system may

reproduce them. Some biases are easy to spot, but many are subtle: what the system treats as 'normal,' which perspectives it foregrounds, what it leaves out, and which moral or political assumptions are built into its default framing. Bias matters in research because it can shape problem framing, interpretation, and which explanations feel 'reasonable.'

Copyright and Intellectual Property

The use of GenAI raises complex copyright questions that differ from traditional plagiarism concerns. While plagiarism is about academic honesty and proper attribution, copyright involves legal rights to use and create content. Two key issues emerge:

1. Training Data: GenAI models are trained on vast amounts of copyrighted material, raising questions about the legality of using their outputs. Investigations have, for example, shown that some LLMs reproduce large segments (word for word) of a Harry Potter book when asked to do so (Ahmed et al. 2026) – an act they have not purchased the rights for.
2. Generated Content: The legal status of AI-generated content remains uncertain. While some jurisdictions require human creativity for copyright protection, others are developing new frameworks to address AI-generated works. Currently, there's ongoing debate about whether and how copyright applies to:
    - Content created entirely by GenAI
    - Content created through human-AI collaboration
    - Derivative works based on AI-generated content

These uncertainties mean researchers should exercise caution when using GenAI-generated content in academic work, particularly in published research or commercial contexts.

References

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